# A crossing change alters two integral Khovanov groups

Disproof of the alternating clause of Problem 00000000620

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**Scope.** The printed conjecture says that every crossing change of an alternating knot perturbs exactly one Khovanov group. We use the standard unreduced, normalized integral groups $\mathrm{Kh}^{i,j}$ of the graded $\mathbb{Z}$-module cube in Bar-Natan's handout [1]. The closed positive two-braid $D = \widehat{\sigma_1^3}$ is a one-component alternating diagram. Change its last crossing to obtain $D' = \widehat{\sigma_1\sigma_1\sigma_1^{-1}}$. We compute both actual cubes directly.

**Theorem 1.** *The groups* $\mathrm{Kh}^{2,5}(D)$ *and* $\mathrm{Kh}^{3,9}(D)$ *are nonzero, whereas* $\mathrm{Kh}^{2,5}(D') = \mathrm{Kh}^{3,9}(D') = 0$. *Thus this single crossing change alters at least two distinct graded groups.*

**The actual diagrams and smoothings.** Number the crossings $r = 0, 1, 2$ from top to bottom. At each crossing the four ports $4r + p$, $p = 0, 1, 2, 3$, are top-left, top-right, bottom-left, bottom-right. The braid strands pair top-left with bottom-right and top-right with bottom-left. Closure arcs join each bottom-left/right to the next top-left/right, cyclically. At a positive crossing the top-left strand passes over the top-right strand; only this choice at $r = 2$ is reversed in $D'$.

An oriented entrance is $(r, b) \in \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}$, and the actual next entrance is $(r + 1, 1 - b)$. This permutation is one six-cycle. For $D$ the over/under status changes at every successor, including the closure. Hence the diagram is a knot and is alternating. Both assertions, the correspondence with the port pairings, and the unique change at $r = 2$ are checked in Lean.

Let $s = (s_0, s_1, s_2) \in \{0, 1\}^3$. At a positive crossing, smoothing 0 is vertical and smoothing 1 joins the two top ports and the two bottom ports horizontally. At the negative crossing these two choices are interchanged. A smoothed circle is an actual connected component of the graph containing the closure pairings and these smoothing pairings. Its representative below is its least port.

| $s$ | circle representatives for $D$ | for $D'$ | $s$ | for $D$ | for $D'$ |
|---|---|---|---|---|---|
| 000 | $0, 1$ | $0$ | 100 | $0$ | $0, 2$ |
| 001 | $0$ | $0, 1$ | 101 | $0, 2$ | $0$ |
| 010 | $0$ | $0, 6$ | 110 | $0, 2$ | $0, 2, 6$ |
| 011 | $0, 6$ | $0$ | 111 | $0, 2, 6$ | $0, 2$ |

These are verified partitions: every graph edge preserves its representative, and each port has a concrete graph path to its representative. The resulting circle type is equivalent to the native graph connected-component quotient.

**The full integer cube.** Use $A = \mathbb{Z}[\epsilon]/(\epsilon^2)$ with $v_+ = 1$, $v_- = \epsilon$, of quantum degrees $1, -1$. Each state contributes $\bigotimes_{\text{circles}} A$. The saddles use the actual multiplication and comultiplication

$$m(1 \otimes 1) = 1, \quad m(1 \otimes \epsilon) = m(\epsilon \otimes 1) = \epsilon, \quad m(\epsilon \otimes \epsilon) = 0,$$

$$\Delta(1) = 1 \otimes \epsilon + \epsilon \otimes 1, \qquad \Delta(\epsilon) = \epsilon \otimes \epsilon.$$

Unaffected circles are matched by their identical port components; the tensor factor reindexing is an actual equivalence. An edge changing $s_r = 0$ to 1 has sign $(-1)^{\sum_{k<r} s_k}$. If a decoration has labels $\ell_c \in \{0, 1\}$, its two normalized degrees are

$$i = |s| - n_-, \qquad j = \sum_c (1 - 2\ell_c) + |s| + n_+ - 2n_-.$$

Here $(n_+, n_-) = (3, 0)$ for $D$, and $(2, 1)$ for $D'$. The complete cube differential preserves $j$, raises $i$ by one, and squares to zero. All these identities are verified on every actual enhanced state. The enhanced-state counts by homological degree are $4, 6, 12, 8$ in degrees $0, 1, 2, 3$ for $D$ and $2, 12, 12, 4$ in degrees $-1, 0, 1, 2$ for $D'$.

**Two non-boundary classes for $D$.** At $(i,j)=(2,5)$ order the six basis vectors as

$$011:(+,-),(-,+); \qquad 101:(+,-),(-,+); \qquad 110:(+,-),(-,+),$$

with circle factors in increasing representative order. The incoming degree-one states are $001, 010, 100$, each labelled $+$; in degree three order the three decorations of 111 with exactly one $+$ by the position of that label. The actual incoming and outgoing matrices are

$$B=\begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix}, \qquad C=\begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 1 & 0 \end{pmatrix}.$$

For $z=(0,-1,1,0,1,0)^t$ and $\lambda=(-1,1,0,0,0,0)$,

$$Cz=0, \qquad \lambda B=0, \qquad \lambda z=-1.$$

Thus $z$ is an integer cycle outside the integer image of $B$; it gives a nonzero class in $\mathrm{Kh}^{2,5}(D)$.

At $(3,9)$ take the 111 state with all three circles labelled $+$. Its outgoing differential is zero. Every incoming split is a $\Delta$ and introduces at least one $-$, so the coordinate functional of this all-$+$ vector vanishes on all boundaries and has value 1 on the vector. Its class in $\mathrm{Kh}^{3,9}(D)$ is therefore nonzero. Lean verifies both probes on the full differential.

**The changed diagram has zero groups in both places.** For $D'$, the top degree $i=2$ has only state 111, with two circles, represented by $0,2$. Let $t_{00},t_{01},t_{10},t_{11}$ be its four labelings, with $0=+$ and $1=-$. In degree one let

$$p_{000}=110:(+,+,+), \quad p_{001}=110:(+,+,-), \quad p_{+}=011:(+), \quad p_{-}=011:(-).$$

The circle order for 110 is $0,2,6$. Direct evaluation of the genuine saddles gives

$$dp_{000}=t_{00}, \qquad d(p_{+}-p_{001})=t_{01}, \qquad dp_{001}=t_{10}, \qquad dp_{-}=t_{11}.$$

These are integer preimages and preserve quantum degree: the four target degrees are $5,3,3,1$. Thus the final differential is surjective in every quantum degree, giving $\mathrm{Kh}^{2,j}(D')=0$ for every $j$. There are no enhanced states in degree 3, so also $\mathrm{Kh}^{3,j}(D')=0$. This proves the theorem.

**Formal objects and identification boundary.** The Lean project constructs the exact port pairings, actual oriented traversal, actual smoothing graph components, native dual-number algebra and tensor-product bases, full signed and doubly graded integer differential, and the groups

$$\mathrm{Kh}^{i,j}=\big(C^{i,j}\cap\ker d\big)/\operatorname{range}\big(d:C^{i-1,j}\longrightarrow C^{i,j}\cap\ker d\big)$$

as native submodules and quotient modules. The two nonzero quotient classes and the changed groups' subsingleton properties exclude additive group isomorphisms. The finite representative tables are certified against actual graph paths.

The identification with the standard unreduced Khovanov construction uses the published diagram/cube definition in [1]. Ambient isotopy of spatial embeddings and the general Reidemeister-invariance theorem are outside this Lean development. Neither trefoil classification nor recognition of $D'$ as an unknot is used in the computation. The conclusion addresses only the printed alternating-knot every-crossing clause; it makes no claim about the separate nonalternating parenthesis.

**Reproduction.** The pinned project is under `lean/`. Run `lean -version`, then `lake build +Main`, `lake env lean -DwarningAsError=true Check.lean`, and `lake env leanchecker -fresh Main`. The evidence records the exact source, all named declarations and generated names, their ordinary axiom audit, and the source-bound execution. Production timings carry no controlled efficiency claim.

# References


[1] D. Bar-Natan, *Khovanov Homology For Knots*, 2022–23 Fast Computations in Knot Theory, author's complete one-page handout. Its $\mathbb{Z}$-module definition, $m,\Delta$, shifts, signs and eight trefoil resolutions are used above.